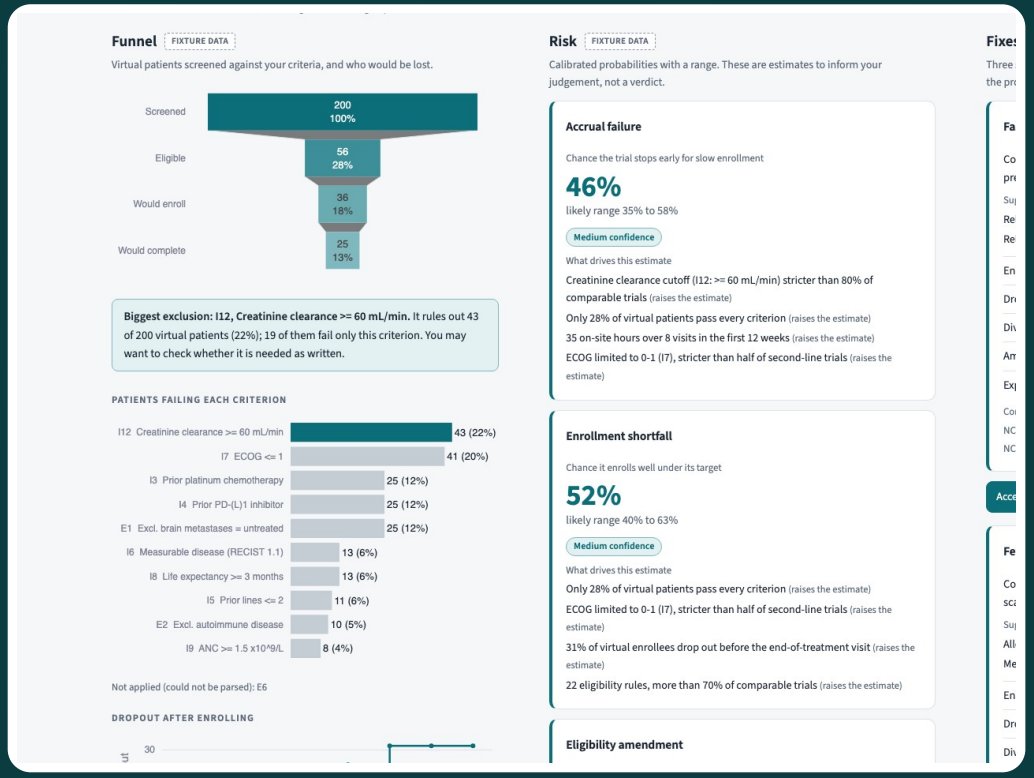
Clinova

Test-fly your NSCLC trial on thousands of virtual patients before enrolling a real one

Every prediction is graded against how real ClinicalTrials.gov trials turned out

https://github.com/RaghulKS/Clinova



THE PROBLEM

Most trial delays are written into the protocol before the first patient is screened

57%

of protocols need a substantial amendment

45% of those are avoidable

$535K

median cost of one Phase III amendment

$141K in Phase II, plus about 3 months

34.5%

of cancer-trial terminations: low accrual, the top reason

22.74% of 8,687 US cancer trials terminated

Clinova pulls the two levers the FDA cost study names: looser enrollment rules and simpler protocols

Sources: docs/ARCHITECTURE.md

WHY NOW

Public version history and LLM parsing make protocol risk checkable today

1

Amendments are public

ClinicalTrials.gov keeps every registered version, so a widened rule is a checkable label

2

LLMs read eligibility

Gemini turns free-text criteria into rules that code can check on virtual patients

3

Design is the AI gap

Tufts CSDD 2024: full AI adoption is lowest in trial design and planning

Sources: docs/ARCHITECTURE.md

WHAT CLINOVA DOES

Clinova test-flies a protocol in five steps, from draft text to a graded risk

1

Read

Draft text or NCT ID to coded rules and visits

2

Simulate

Virtual NSCLC cohort, screened in code; persona agents decide

3

Calibrate

LightGBM on real trials, explained by SHAP

4

Optimize

Test edits; trade-offs with ranges and dollars

5

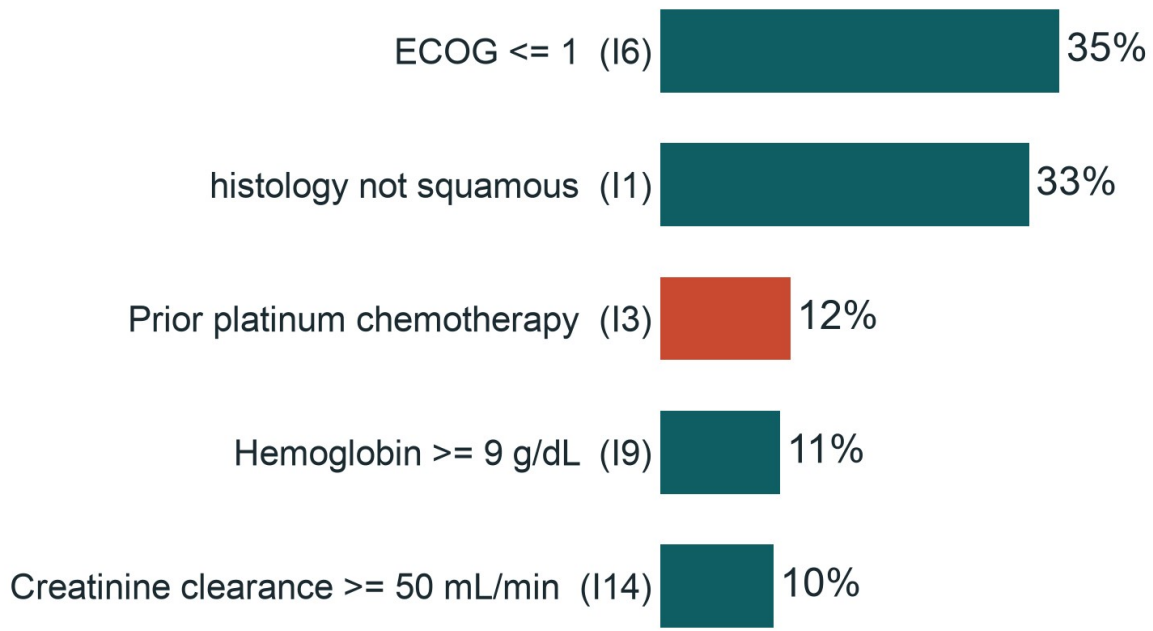
Learn

Log every prediction, score it when the trial ends

Sources: docs/ARCHITECTURE.md

DRAFT: placeholder numbers

Backtest: from NCT03232892's first protocol, Clinova finds where enrollment stalls



NCT03232892

Terminated 2019-09-11: “Low Accrual”

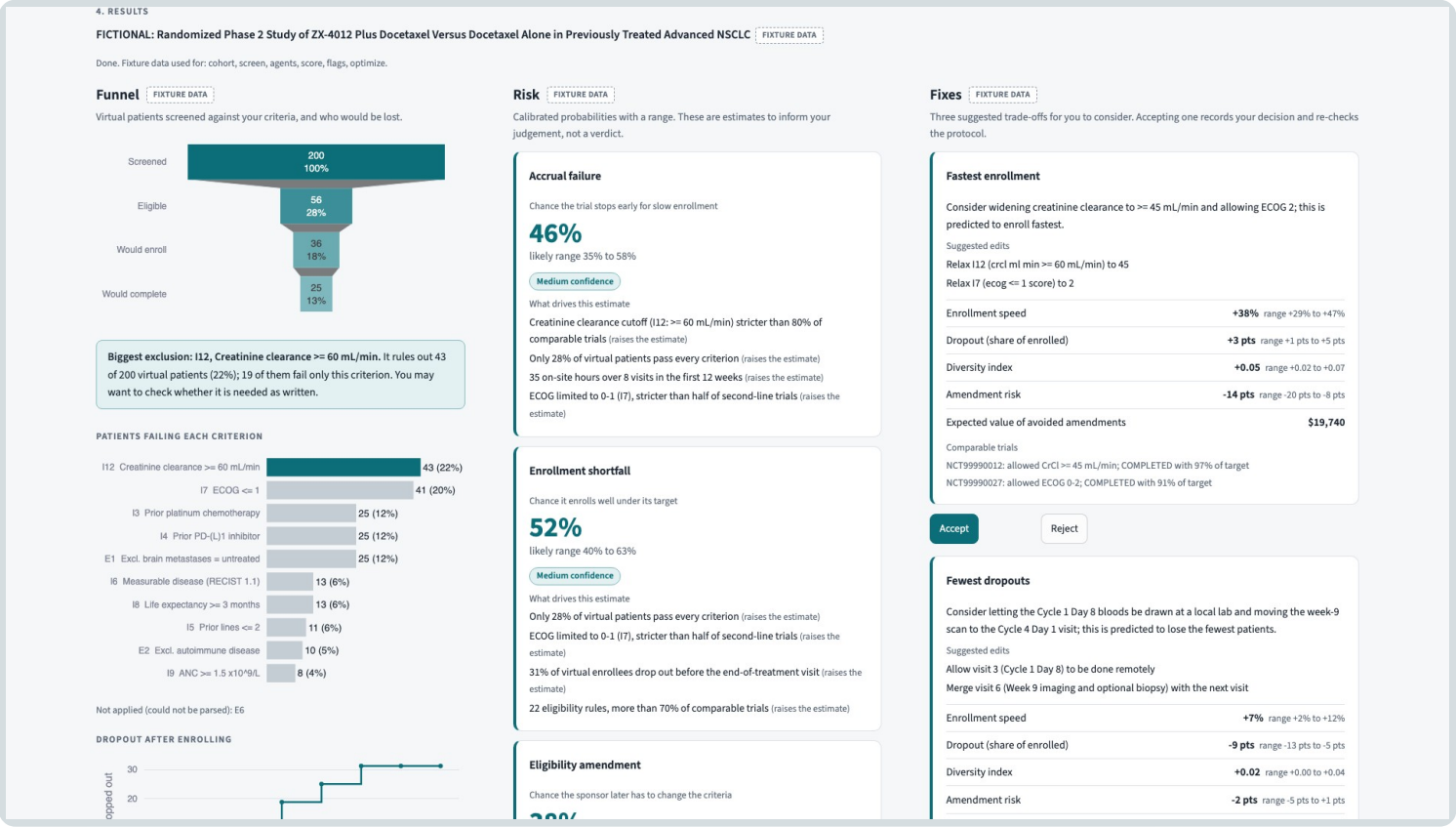
- 21.4% of 1,000 virtual patients pass the first registered criteria
- ECOG <= 1 excludes the most (35%)
- Did our top 3 flags include the widened rule? [TBD: demo flags (docs/deck/data/demo.json)]

Coral bars: rules the sponsor later widened

Sources: docs/demo_trial.md, docs/gate1_e2e.json (seed 1), docs/deck/data/demo.json

LIVE DEMO

One screen shows the funnel, the calibrated risk and three fixes



- Funnel: who is excluded by which rule, who enrolls, who completes
- Risk: calibrated probability with a range and top drivers in plain words
- Fixes: three trade-offs; accept one and it re-runs in seconds

Sources: docs/screens/03-results.png

OPTIMIZER

DRAFT: placeholder numbers

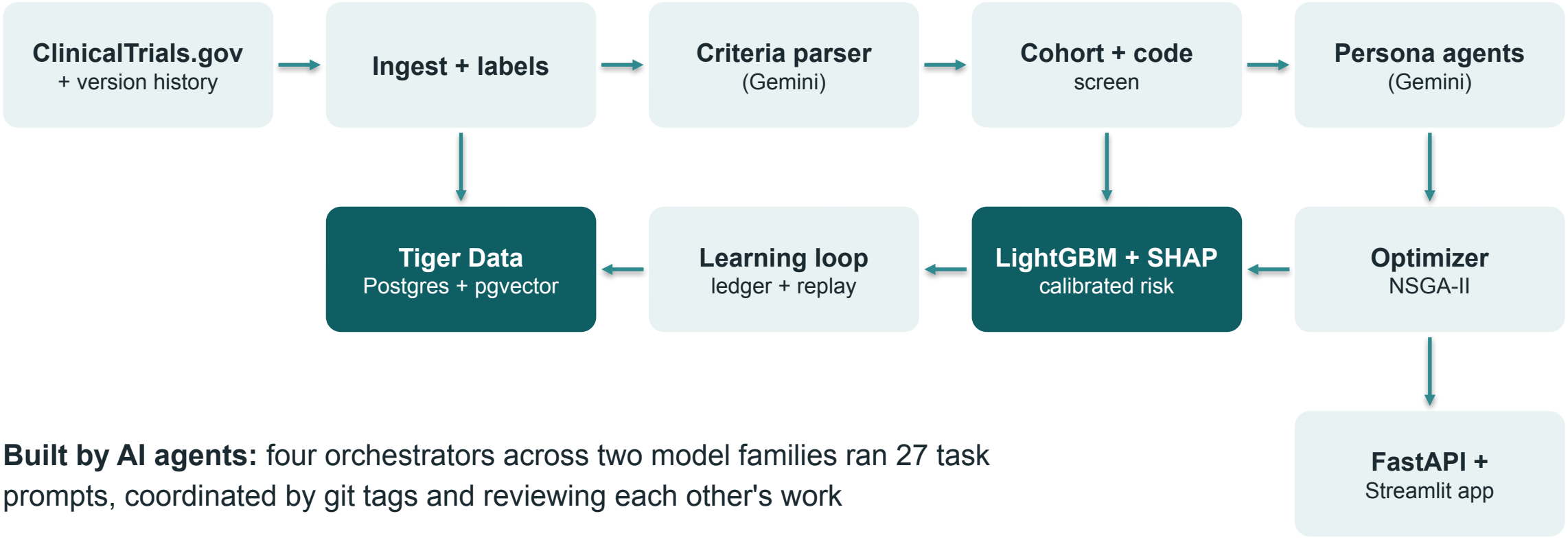
The optimizer proposes three trade-offs, each with a range and a dollar value

[TBD: demo fixes from docs/deck/data/demo.json]

Sources: docs/deck/data/demo.json

ARCHITECTURE

Small, tested modules around one Tiger Data store, built entirely by AI agents

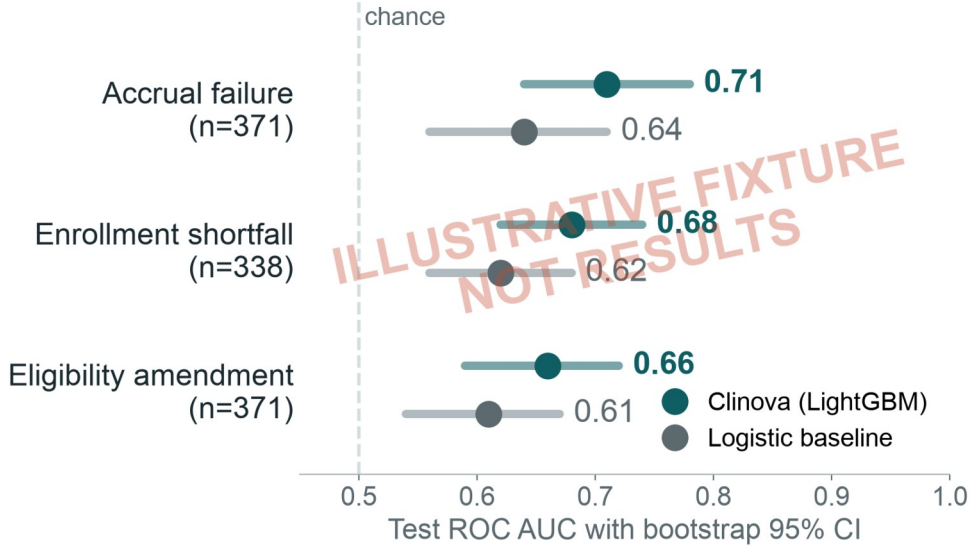


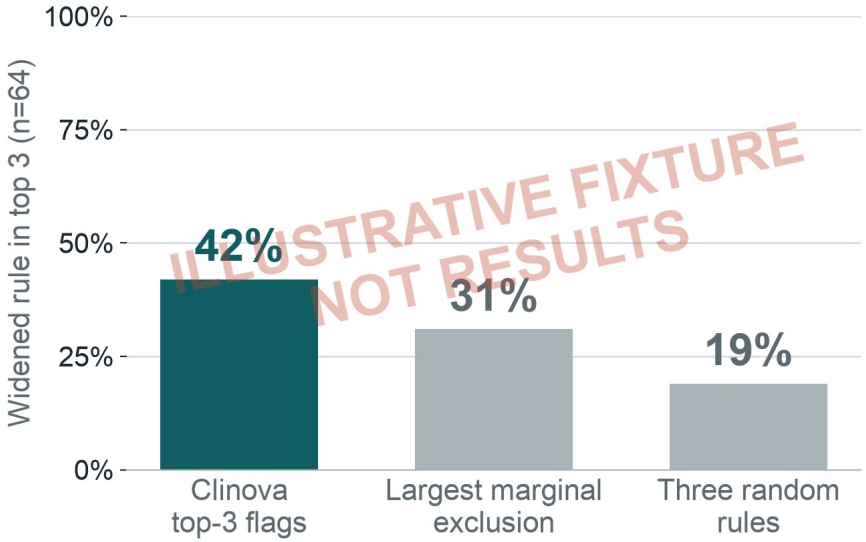
Built by AI agents: four orchestrators across two model families ran 27 task prompts, coordinated by git tags and reviewing each other's work

Sources: docs/ARCHITECTURE.md, orchestration/tasks/

DRAFT: placeholder numbers

On unseen trials, Clinova's accrual-failure AUC is [TBD: metrics.json accrual_fail AUC] vs [TBD: metrics.json accrual_fail AUC baseline] for a baseline



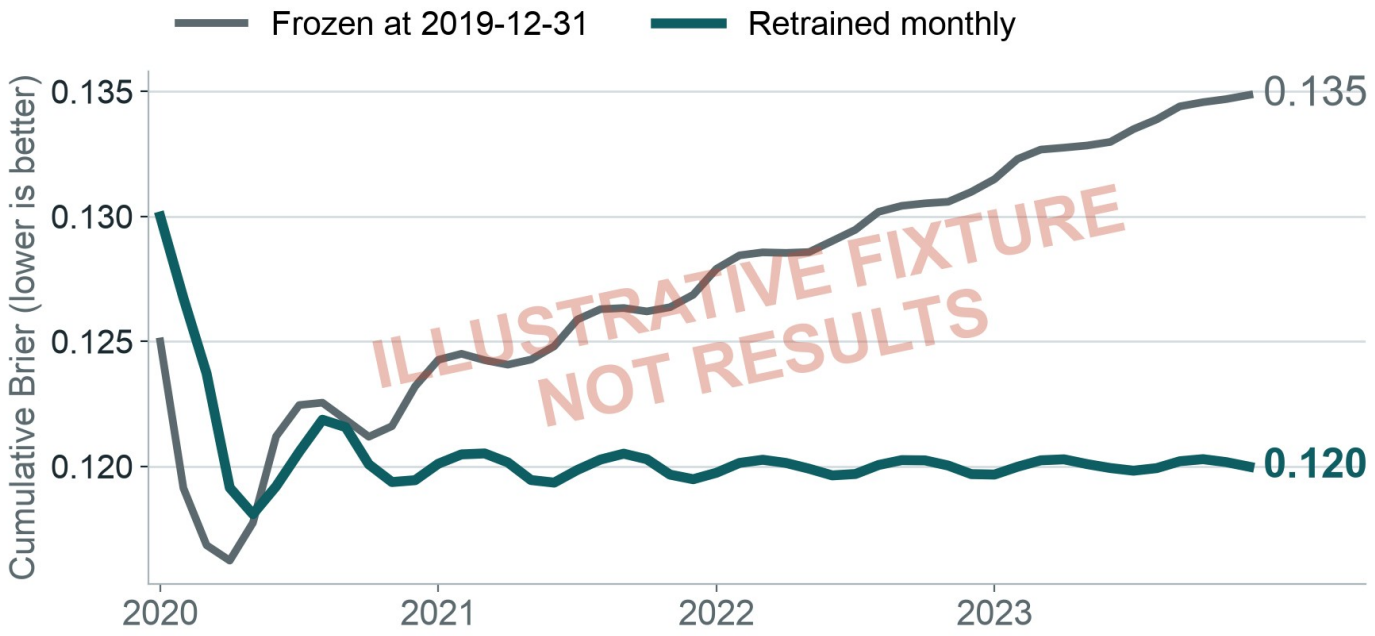


- Accrual failure: AUC [TBD: metrics.json accrual_fail AUC] (95% CI) vs baseline [TBD: metrics.json accrual_fail AUC baseline] ()
- Hit rate: [TBD: metrics.json hit_rate] (random [TBD: metrics.json hit_rate random], largest exclusion [TBD: metrics.json hit_rate marginal])
- Simulation ablation: [TBD: metrics.json ablation diff]

Sources: docs/metrics.json

DRAFT: placeholder numbers

The learning loop grades every prediction; retrained Brier [TBD: replay.json retrained Brier] vs frozen [TBD: replay.json frozen Brier]



- Replay [TBD: first month] to [TBD: last month]: score each month, then learn its outcomes
- Pooled Brier over [TBD: replay.json windows] scored windows: retrained [TBD: replay.json retrained Brier], frozen [TBD: replay.json frozen Brier]
- Every live prediction is logged in Tiger Data and scored when the trial ends

Sources: docs/replay.json

LIMITATIONS AND RESPONSIBLE USE

We show what the evidence can and cannot support

Registry edits are a proxy

A widened rule on ClinicalTrials.gov is public evidence of an amendment, not the same thing

Stop reasons are free text

Some terminations are business decisions; labels were spot-checked by a second model

Personas are synthetic

General-population records with clinical traits from published distributions, labelled as assumptions

Predictions are correlational

Risk scores rank protocols; they do not prove that a rule caused a failure

Clinova advises; the author decides. No real patient data. Every decision is logged.

Sources: docs/ARCHITECTURE.md, docs/metrics.json

ROADMAP

Next: more indications, site-level simulation and structured protocol input

1. Lymphoma, ophthalmology and immunology
2. Site-level enrollment simulation
3. Persona model tuned on expert feedback
4. Structured protocol input (ICH M11)
5. Full GxP validation package

Run it yourself

github.com/RaghulKS/Clinova

MIT license. docker compose up, then open the app; demo mode needs no API keys.

Sources: docs/ARCHITECTURE.md